In a `tzap` `channels.conf` file, each line corresponds to a specific TV channel, and the fields represent different technical parameters required to tune into that channel. Let's break down the components of the example line you provided:

SVT2 HD(com hem):314000000:INVERSION_AUTO:6875000:FEC_NONE:QAM_256:4097:4609:1046

1. Channel Name (`SVT2 HD(com hem)`)

- **SVT2 HD**: This is the name of the channel. It's a string that will be displayed in the TV interface. The name can be any label you wish, typically the channel's name or the network's name.
- **(com hem)**: This is an optional label for the provider or service (e.g., a specific cable operator or satellite service), and in this case, it indicates that the channel is provided by `Com Hem`.

2. Frequency (`314000000`)

- This is the frequency at which the channel is broadcast, in **Hz** (Hertz).
- `314000000` Hz = 314 MHz, which is a typical cable frequency.

3. Inversion (`INVERSION_AUTO`)

- This specifies whether the symbol rate (modulation) is inverted or not.
- **INVERSION_AUTO**: This means that the inversion is automatically detected by the tuner. It could also be set to `INVERSION_ON` or `INVERSION_OFF` in some cases, depending on the signal conditions.

4. Symbol Rate (`6875000`)

- The symbol rate refers to how many symbols are transmitted per second. It is given in **symbols per second** (or **baud**).
- `6875000` is the symbol rate, which is 6.875 MHz (commonly used for cable TV).

5. Forward Error Correction (FEC) (`FEC_NONE`)

- FEC is used to correct errors that occur during transmission.
- **FEC_NONE** means that no forward error correction is used. Other values might be `FEC_1_2`, `FEC_2_3`, `FEC_3_4`, etc., representing different levels of error correction.

6. Modulation (`QAM_256`)

- This specifies the modulation type used to encode the signal.
- **QAM_256** indicates that 256-QAM (Quadrature Amplitude Modulation) is used, which is typical for cable TV services. The number refers to the number of different signal states that can be used.

7. Service ID (`4097`)

- This is the service ID for the channel, used to uniquely identify it within the DVB or cable network.
- `4097` is the service ID in this case.

8. Network ID (`4609`)

- The network ID uniquely identifies the network the channel belongs to.
- `4609` refers to the specific cable network the channel is a part of.

9. Transport Stream ID (`1046`)

- This is the transport stream ID, which uniquely identifies the stream of data that is sent to the receiver.
- `1046` is the stream ID in this example.

Recap of the fields:

- **Channel Name**: `SVT2 HD(com hem)`
- **Frequency**: `314000000` Hz (314 MHz)

- **Inversion**: `INVERSION_AUTO`
- **Symbol Rate**: `6875000` symbols per second (6.875 MHz)
- **FEC**: `FEC_NONE`
- **Modulation**: `QAM_256`
- **Service ID**: `4097`
- **Network ID**: `4609`
- **Transport Stream ID**: `1046`

Each of these fields is important for configuring the tuner to properly receive and decode the broadcast signal.